



Exam #2 Study Guide

15 True/False questions (1 point each) + 5 Numerical Free Response questions (5 points each)

75 Minutes, 40 Points

The listed topics are for numerical Qs. Conceptual Qs will show up from any post-Exam 1 topic.

Multiple Linear Regression (1 Q):

- Coefficient Estimation
- Residuals, MSE, SST, SSE, SSReg, R^2 , Adjusted R^2 relationships
- Prediction/Interpretation w/Binary, Categorical, Interaction terms

MLR Inference (1 Q):

- # of Predictors counting
- Parameter standard errors, MLR Confidence & Prediction intervals
- t -Test, Hypothesis Testing & Conclusions
- ANOVA Table
- F-test and nested F-test

Model Diagnostics (1 Q):

- Leverage, Standardized residuals, Outliers
- Cook's distance
- Variance inflation factor

Model Selection (1 Q):

- Automatic variable selection – Forward, Backward, Best
- Adjusted R^2 , Mallows' C_p /AIC, BIC
- k-fold Cross-validation, LOOCV

1 additional numerical Q from any of the sections

Vocabulary to Know:

- Total Sum of Squares vs. Error Sum of Squares (aka. Residual Sum of Squares) vs. Regression Sum of Squares and relationship with Residual, Mean Squared Error, Residual Standard Error, R^2 , adjusted R^2
- Mean Squared Error vs. Residual Standard Error
- Variance vs. Standard Error
- p-value vs. critical α vs. t-statistic and relationship w/hypothesis testing and conclusion
- Confidence vs. Prediction interval
- Nested models meaning
- Explanatory variables/parameters needed vs. total coefficients/parameters



- Forward vs. backward stepwise selection
- Mallows's C_p = AIC in multiple linear regression
- Out-of-sample validation vs. k-folds cross-validation vs. LOOCV
- Linear model violations vs. issues
- Heteroscedasticity vs. homoscedasticity
- Outlier vs. high leverage vs high influence (Cook's Distance)
- Collinearity (none vs. mild vs. severe), Variance inflation factor

Formulas to Remember:

- $SST, SSE, SSReg, s_y^2, R^2, t(b_j)$ from first exam
- ANOVA Table relationships
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- $R_a^2 = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - (1 - R^2) \left(\frac{n-1}{n-(p+1)} \right)$
- Whole $F = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{\frac{R^2}{1-R^2}}{\frac{p}{n-(p+1)}}$
- Nested $F = \frac{\frac{(SSE_0 - SSE_{full})}{q}}{\frac{SSE_{full}}{n-(p+1)}} = \frac{\frac{R_{full}^2 - R_{reduced}^2}{1-R_{full}^2}}{\frac{q}{n-(p+1)}}$
- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (X'X)^{-1}}$
- Confidence Interval: $b_j \pm t_{critical} * se(b_j)$
- $r_{partial} = \frac{t_i}{\sqrt{t_i^2 + n-(p+1)}}$
- Breusch-Pagan: Reject H_0 if $\frac{SSReg_{aux}}{2} > \chi_{\alpha,p}^2$
- Standardized Residual $e_{std} = \frac{e_i}{se(e_i)} = \frac{y_i - \hat{y}_i}{s\sqrt{1-h_i}}$
- SLR Leverage $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2}$
- Cook's Distance $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}} \right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- $VIF_j = \frac{1}{1-R_j^2}$
- Mallows's $C_p = \frac{1}{n} (SSE_p + 2ps_{full}^2)$
- $BIC = \frac{1}{n} (SSE_p + \ln(n) ps_{full}^2)$
- $CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$
- LOOCV: $CV = \frac{1}{n} \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$